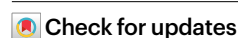


Self-reported side effects of semaglutide and tirzepatide in online communities

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Social media can reveal patient experiences with glucagon-like peptide-1 receptor agonists (GLP-1 RAs) that extend beyond clinical trial data. We analysed 410,198 Reddit posts (May 2019–June 2025) mentioning semaglutide or tirzepatide. A total of 67,008 users self-reported using these medications, and 43.5% described at least one side effect. Gastrointestinal symptoms predominated, including nausea (36.9%), fatigue (16.7%), vomiting (16.3%), constipation (15.3%) and diarrhoea (12.6%). Notably, reproductive symptoms (for example, menstrual irregularities) and temperature-related complaints (for example, chills and hot flushes) emerged as unrecognized potential effects. These findings highlight patient concerns not well captured in current labelling or trials. Large-scale social media analysis can complement traditional pharmacovigilance by detecting emerging safety signals and expanding understanding of the real-world safety profile of GLP-1 RAs.

Two glucagon-like peptide-1 receptor agonists (GLP-1 RAs), semaglutide and tirzepatide, have rapidly gained popularity for type 2 diabetes management and weight control. Clinical trials and regulatory documents have established gastrointestinal symptoms, such as nausea, vomiting and diarrhoea, as common side effects of these medications. Social media data provide the opportunity to capture a broader range of patient experiences, including symptoms not formally recognized in drug labelling¹. Understanding these real-world reports can help clinicians and regulators build a more comprehensive safety profile.

In this paper, we systematically characterize self-reported side effects of semaglutide and tirzepatide shared on Reddit, a social media platform with over 100 million daily active users anonymously posting in topical forums (subreddits)². Reddit was selected as the data source for this study due to its large, publicly accessible, topic-specific health communities and its long-form, context-rich user discussions. Prior work has established Reddit as a valuable source for studying health-related disclosures, including sensitive or stigmatized medical experiences, and as an important resource for public health and computational social science research^{3,4}. Unlike other major platforms, which provide either restricted, unreliable or prohibitively

costly application programming interface access, publicly released researcher-curated corpora have made Reddit data practically feasible for large-scale public health research.

We conducted a cross-sectional study of self-reported side effects of semaglutide and tirzepatide using Reddit data from May 2019 through June 2025. Posts and comments were collected from nine large subreddits that discuss GLP-1 RAs or weight management. We first identified posts where users disclosed taking semaglutide or tirzepatide in any Food and Drug Administration-approved formulation for type 2 diabetes or weight management using a generative pre-trained transformer (GPT)-4o-mini-based classifier. This classifier also extracted the specific medications mentioned. Among posts indicating self-use, we applied a GPT-4.1-mini-based classifier to extract self-reported side effects and map them to Medical Dictionary for Regulatory Activities (MedDRA) Preferred Terms (PTs). Across the May 2019–June 2025 period during which eligible posts appeared, 172,679 of 410,198 Reddit posts mentioning semaglutide or tirzepatide indicated personal use by 67,008 users. Among them, 29,172 users (43.5%) reported at least one side effect, with a mean of 2.7 (standard deviation 2.4) PTs per user.

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Table 1 presents the prevalence of PTs among users reporting side effects. Nausea was most common (36.9%), followed by fatigue (16.7%), vomiting (16.3%), constipation (15.3%) and diarrhoea (12.6%). Other symptoms reported by $\geq 5\%$ of users who disclosed side effects included decreased appetite, abdominal pain, gastroesophageal reflux, headache, abdominal distension and dizziness. The most frequent co-occurring PTs were nausea and vomiting, with 2,917 users (10.0%) self-reporting both (Supplementary Table 1).

Overall, 65.8% of semaglutide and/or tirzepatide users with side effects reported at least one gastrointestinal symptom. Psychiatric symptoms were noted by 12.9%, with anxiety (4.2%) and depression (2.8%) being most common.

As a secondary analysis, we examined monthly frequencies of the ten most commonly reported MedDRA PTs. Symptom counts increased in parallel with overall discussion volume but did not show distinct or interpretable temporal shifts (Supplementary Fig. 1).

As an exploratory analysis, we conducted a subanalysis of users who exclusively mentioned one formulation to determine differential side effect rates. Of the 29,172 users who self-disclosed at least one side effect, 17,937 (61.5%) exclusively mentioned semaglutide formulations and 7,125 (24.4%) exclusively mentioned tirzepatide formulations.

We then examined side effect frequencies separately for users who mentioned semaglutide exclusively ($n = 17,937$) versus tirzepatide ($n = 7,125$). Among semaglutide users reporting side effects, the most commonly reported symptoms were nausea (39.4%), fatigue (16.1%), vomiting (18.0%), constipation (14.9%) and diarrhoea (12.4%). Among tirzepatide users reporting side effects, the most commonly reported symptoms were nausea (28.6%), fatigue (14.7%), vomiting (11.1%), constipation (12.9%) and diarrhoea (12.5%). Injection site reactions, myalgia, insomnia and temperature-related symptoms (chills and feeling cold) were each reported by 1–4% of tirzepatide users. A complete descriptive summary for each formulation is provided in Supplementary Table 2.

In this large-scale Reddit analysis, over 67,000 semaglutide or tirzepatide users were identified, with 44% reporting side effects. Consistent with the findings of clinical trials, gastrointestinal symptoms were reported most commonly by these users^{5–8}. The most frequently reported gastrointestinal effects were nausea, vomiting, constipation, diarrhoea and abdominal pain, which are the only adverse reactions featured as occurring more commonly than with placebo in the Food and Drug Administration-approved prescribing information for all five versions of these medications. Additional symptoms reported by $\geq 5\%$ of users (for example, decreased appetite, eructation, distension and dizziness) have similarly been noted in trials as more common than placebo^{5–8}. Headache, though common in trials, shows mixed placebo comparisons, while fatigue, Reddit's second most reported symptom, has met reporting thresholds in relatively few trials^{5–8}.

We also identified several side effects not previously reported with semaglutide or tirzepatide. Nearly 4% of Reddit users with side effects reported reproductive symptoms, notably menstrual changes like intermenstrual bleeding (0.9%), heavy bleeding (0.9%) and irregular cycles (0.7%). These rates would be higher in female-only samples. While weight loss may alleviate some reproductive issues, both weight gain and loss are linked to menstrual irregularities in women with obesity⁹. Further, GLP-1 RAs are thought to impact food intake, in part, by engaging receptors in the hypothalamus, a brain region that also plays a central role in menstrual cycle regulation¹⁰. Additionally, some users described symptoms possibly linked to altered temperature regulation (for example, chills, hot flushes and pyrexia), warranting further investigation in placebo-controlled trials or post-market data, particularly due to glucagon's associations with increased thermogenesis¹⁰.

Strengths of our study include a large and contemporary sample, the focus on unprompted, self-disclosed experiences, the systematic classification of symptoms using MedDRA and the development of a

straightforward large language model-based pipeline to detect emerging, potentially under-reported signals in a timely manner.

Several limitations should be noted. First, patients using weight loss subreddits may differ from the overall population prescribed GLP-1 RAs. Specifically, Reddit users tend to be younger, are more likely to be male and are disproportionately located in the USA^{11,12}. The demographic composition of Reddit users and self-selection into health-related discussions limit the generalizability of our findings. Without cross-platform validation or comparison to structured pharmacovigilance databases, we cannot quantify the magnitude or direction of these biases. Although Reddit provides one of the largest and most active online health communities and remains among the few platforms with accessible, high-volume data, future work should extend this approach to other sources—such as specialized patient forums, clinical review platforms or any social media systems that restore stable application programming interface access—to assess the consistency of patient-reported symptom patterns across environments. Our results should be interpreted as hypothesis-generating signals that require confirmation through traditional surveillance systems and prospective studies in representative populations¹³.

Second, because users were not prompted to disclose all side effects, we cannot estimate their true prevalence. Participation in health-related subreddits is voluntary, and individuals who experience more severe, persistent or distressing symptoms may be more likely to post about them than those with neutral or positive experiences. Even those who reported side effects may not have disclosed every symptom experienced, and individuals' beliefs about which symptoms are potentially attributable to the medication may have influenced their report. Further, earlier work has found that, while clinical trials often rely on self-reporting, gastrointestinal symptoms are often under-reported and inconsistently reported compared with validated measures for functional gastrointestinal disorders^{14,15}. For these reasons, self-reported experiences cannot be used to determine true event frequency or causality. The symptoms discussed here may be attributable to weight loss itself, changes in comorbidities or other unrelated factors. Additionally, dosage, route of administration and treatment duration were inconsistently reported in posts and were not analysed. Our analysis prioritized breadth of symptom identification over temporal granularity. While we examined monthly trends in common symptoms (Supplementary Fig. 1), detailed assessment of symptom onset timing or dose–response relationships would require structured metadata (for example, prescription dates or dose titration schedules) rarely disclosed in social media posts. Examination of these variables would require data sources with more complete and structured medication histories, such as electronic health records or prospective registries. We were unable to determine when or how frequently these events occurred over the course of treatment or whether event occurrence depended on patient characteristics, such as the presence of type 2 diabetes. Results should be viewed as hypothesis-generating signals about patient-perceived symptoms rather than estimates of causal effects or true event rates.

Third, natural language processing can misclassify or overlook nuanced context, and adverse drug event extraction from social media texts remains an active area of research¹⁶. We validated the specific models deployed in this study against manually annotated samples and achieved high performance comparable to prior adverse event extraction systems^{1,16}. While systematic benchmarking across large language model architectures represents an important direction for future methods research, our validation approach ensured reliable extraction for this application. Additionally, certain self-reported effects may not fit neatly into existing MedDRA concepts or may have been misclassified based on the user's description. For example, some events classified as 'gastroesophageal reflux' based on user terminology might have instead been recorded as 'dyspepsia' in the context of a clinical trial. In our exploratory subanalysis restricted to users whose

Table 1 | Frequency of MedDRA PTs among 29,172 Reddit users taking GLP-1 RAs

MedDRA term	No. of users self-reporting	Percentage of users with self-disclosed PT
Gastrointestinal disorders	19,185	65.8
Nausea	10,764	36.9
Vomiting	4,749	16.3
Constipation	4,463	15.3
Diarrhoea	3,686	12.6
Abdominal pain	2,493	8.5
Eructation	2,010	6.9
Gastroesophageal reflux disease	1,877	6.4
Abdominal distension	1,488	5.1
Flatulence	1,055	3.6
Abdominal discomfort	900	3.1
Gastrointestinal disorder	883	3.0
Dyspepsia	662	2.3
Retching	350	1.2
Dry mouth	321	1.1
Breath odour	187	0.6
Gastrointestinal motility disorder	167	0.6
Impaired gastric emptying	167	0.6
Dysphagia	142	0.5
General disorders and administration site conditions	8,395	28.8
Fatigue	4,870	16.7
Asthenia	738	2.5
Early satiety	666	2.3
Pain	619	2.1
Malaise	589	2.0
Chills	332	1.1
Injection site pain	283	1.0
Hunger	220	0.8
Feeling cold	183	0.6
Illness	156	0.5
Thirst	147	0.5
Injection site reaction	144	0.5
Pyrexia	139	0.5
Nervous system disorders	5,704	19.6
Headache	1,784	6.1
Dizziness	1,445	5.0
Somnolence	492	1.7
Dysgeusia	398	1.4
Syncope	346	1.2
Tremor	333	1.1
Lethargy	311	1.1
Paraesthesia	301	1.0
Migraine	294	1.0
Cognitive disorder	267	0.9
Burning sensation	190	0.7
Taste disorder	177	0.6
Metabolism and nutrition disorders	5,205	17.8
Decreased appetite	3,371	11.6
Hypoglycaemia	600	2.1
Increased appetite	549	1.9
Dehydration	473	1.6
Food aversion	365	1.3
Food craving	171	0.6

Table 1 (continued) | Frequency of MedDRA PTs among 29,172 Reddit users taking GLP-1 RAs

MedDRA term	No. of users self-reporting	Percentage of users with self-disclosed PT
Psychiatric disorders	3,756	12.9
Anxiety	1,230	4.2
Insomnia	901	3.1
Depression	827	2.8
Panic attack	280	1.0
Irritability	257	0.9
Anhedonia	191	0.7
Libido decreased	161	0.6
Mood swings	159	0.5
Apathy	149	0.5
Sleep disorder	148	0.5
Depressed mood	142	0.5
Skin and subcutaneous tissue disorders	2,552	8.7
Alopecia	914	3.1
Hyperhidrosis	362	1.2
Pruritus	335	1.1
Rash	193	0.7
Acne	183	0.6
Dermatitis allergic	166	0.6
Urticaria	149	0.5
Musculoskeletal and connective tissue disorders	1,833	6.3
Myalgia	540	1.9
Arthralgia	336	1.2
Back pain	325	1.1
Muscle atrophy	194	0.7
Muscle spasms	158	0.5
Musculoskeletal pain	134	0.5
Reproductive system and breast disorders	1,142	3.8
Intermenstrual bleeding	266	0.9
Heavy menstrual bleeding	261	0.9
Menstruation irregular	211	0.7
Menstrual disorder	178	0.6
Dysmenorrhoea	168	0.6
Investigations	832	2.9
Weight increased	415	1.4
Cardiac disorders	718	2.5
Tachycardia	512	1.8
Palpitations	205	0.7
Respiratory, thoracic and mediastinal disorders	673	2.3
Dyspnoea	169	0.6
Vascular disorders	628	2.2
Hypotension	166	0.6
Hot flush	151	0.5
Infections and infestations	453	1.6
Eye disorders	388	1.3
Visual impairment	218	0.7
Renal and urinary disorders	357	1.2
Injury, poisoning and procedural complications	234	0.8
Contusion	157	0.5

PTs are grouped by primary system organ classes (bold) in descending order. PTs reported by less than 0.5% of users and system organ classes reported by less than 1.0% are omitted for brevity. See Supplementary Table 3 for PTs ordered purely by frequency and Supplementary Tables 4 and 5 for frequencies of corresponding MedDRA high-level terms and high-level group terms.

posts exclusively mentioned one formulation, we present descriptive comparisons of side effect frequencies but do not make formal statistical inferences about medication-specific effects. Without demographic and clinical data to match or adjust for confounding between medication groups, we cannot determine whether observed differences in complaint frequencies are attributable to the formulations themselves or to differences in the populations using each formulation. Such analyses would require data sources with detailed patient characteristics to enable appropriate comparison groups. Future work should stratify symptom patterns by medication using data sources with richer clinical context and validated methods for establishing medication-specific causal effects.

Despite these limitations, social media data provide important insights into symptoms that are of concern to semaglutide and tirzepatide users and may be helpful in identifying rarer or under-reported effects that warrant further investigation. Online health communities are increasingly influential in shaping patient expectations and treatment decisions, and a growing body of research examines these platforms across a range of contexts^{17–21}. Future work could extend this framework by examining more detailed temporal dynamics in symptom patterns and how they vary by medication type or formulation; incorporating severity distinctions in patient-reported symptoms; and evaluating multilingual or non-Reddit digital communities to understand cross-platform and cross-cultural variation in patient experiences.

Capturing these patient-reported experiences can alert clinicians and researchers to emerging patient concerns related to GLP-1 RA use. Our data suggest that clinicians may encounter patient inquiries about menstrual irregularities and body temperature fluctuations in addition to commonly reported gastrointestinal symptoms. We recommend that future prospective trials and post-market surveillance studies systematically assess fatigue and reproductive symptoms, particularly menstrual changes in female participants, using validated instruments rather than relying solely on passive adverse event reporting. These findings may also inform updates to patient counselling materials and regulatory labelling to address symptoms of concern better to patients in real-world settings. We encourage researchers to examine whether these patient-reported concerns occur more frequently with semaglutide and/or tirzepatide than with placebo.

Methods

Posts and comments were collected from nine large subreddits (forums) that discuss GLP-1 RAs or weight management. These subreddits were selected to align with those analysed by prior researchers who estimated real-world weight loss outcomes of these medications²². Relevant posts and comments containing mentions of semaglutide or tirzepatide were identified using a list of brand and generic names and common misspellings. Data were collected for the period 1 January 2015 to 30 June 2025; in practice, all posts that met inclusion criteria were dated May 2019–June 2025. All data were collected using Pushshift and Arctic Shift (an extension of Pushshift)^{23,24}. Please see the Supplementary Methods for subreddits used and a full list of search terms.

We first identified posts in which users personally disclosed taking either semaglutide or tirzepatide in any formulation currently approved by the Food and Drug Administration for the treatment of type 2 diabetes or weight management using a GPT-4o-mini based text classifier (prompt in Supplementary Methods). The classifier also extracted the specific medications being taken. The classifier's self-use performance was evaluated against a manually annotated set of 100 randomly selected posts and had a sensitivity of 86% and specificity of 96%. Only posts that met the self-disclosure criterion were included in subsequent analyses.

We evaluated the model's medication extraction accuracy on a random sample of 100 posts containing 109 medication mentions. The model achieved a precision of 95% and recall of 98% (F1 = 96%) at

the medication level. Common errors included extracting a specific medication when the post did not clearly state which drug was being used, often inferring the medication from the subreddit name (for example, assuming Ozempic for posts in r/Ozempic) rather than from explicit mentions in the post content ($n = 4$), and missing secondary (non-semaglutide or non-tirzepatide) medications ($n = 2$). At the post level, 93.0% of posts had all medications correctly extracted.

Among posts indicating self-use, we applied a second classifier, based on the GPT-4.1-mini model, to extract self-reported side effects. Prior work has shown that both classical and transformer-based natural language processing systems can map free text to standardized vocabularies such as MedDRA^{25–27}. Our pipeline follows this normalization paradigm but uses a large language model classifier to directly map side effects mentioned in the user-generated text to standardized MedDRA PTs using retrieval augmented generation (prompt in Supplementary Methods). Manual evaluation of 100 randomly selected posts indicated that the classifier had a sensitivity (recall) of 97% and a positive predictive value (precision) of 87% for identifying the presence or absence of individual side effects. Because true negatives cannot be enumerated meaningfully for named entity recognition models at the span level, specificity is undefined for this granularity. The classifier extracted 2,013 unique side effect phrases, which were matched to existing MedDRA PTs via exact matching and manual review when needed.

Percentages were calculated based on the population of users who disclosed at least one side effect. Because users were not prompted to disclose side effects, we cannot estimate their true prevalence. Further, uncontrolled self-reported experiences cannot be used to determine causality. The symptoms discussed here may be attributable to weight loss itself, changes in comorbidities or other unrelated factors.

For users mentioning one formulation exclusively (either semaglutide or tirzepatide but not both), we conducted exploratory descriptive comparisons of side effect frequencies. For each MedDRA PT, we calculated proportions in each medication group to characterize the distribution of reported symptoms. These comparisons are presented descriptively without formal statistical testing, as we lack the demographic and clinical data necessary to adjust for potential confounding factors that may differ between users of each medication.

GPT-4o-mini was chosen for self-disclosure classification because, at the time that component of the study was conducted, it was among the most capable lightweight models available for large-scale processing at feasible cost. Midway through the project, OpenAI released GPT-4.1-mini with documented performance improvements over GPT-4o-mini across a range of tasks including instruction following (<https://openai.com/index/gpt-4-1/>), and we therefore used GPT-4.1-mini for identifying mentions of side effects and mapping them to MedDRA terms. Larger, reasoning and non-mini models (for example, GPT-4o or GPT-4.1) were not employed for cost reasons.

All analyses were conducted in Python 3.11 using standard data processing libraries. Data were collected from publicly accessible Reddit posts and comments using Pushshift and Arctic Shift during the study period. No attempts were made to access private or password-protected content. As data were public and non-identifiable, institutional review board review was not required. Human Ethics and Consent to Participate declarations were not applicable.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this study are publicly available from Reddit.com or are accessible via Pushshift and Arctic Shift. Due to Reddit's Terms of Use, we are unable to share the raw data. All Reddit data in this study were used in accordance of Reddit's Terms of Use.

Code availability

Code to reproduce the analyses can be found at <https://github.com/sehgal-neil/glp1-side-effects-analysis>.

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Author contributions

N.K.R.S.: conceptualization, methodology, formal analysis, investigation, data curation and writing—original draft. J.S.T.: conceptualization, methodology and writing—review and editing. L.U.: conceptualization, methodology and writing—review and editing. S.C.G.: conceptualization, methodology, writing—review and editing, and supervision.

Competing interests

J.S.T. reports receiving an investigator-initiated grant, on behalf of the University of Pennsylvania, from Novo Nordisk and receiving consulting fees from Currax Pharmaceuticals, LLC. The other authors declare no competing interests.

Additional information

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Self-reported side effects of semaglutide and tirzepatide in online communities

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APPENDIX

Figure 1. Monthly Mentions of the Ten Most Common MedDRA Preferred Terms (2019–2025)

Number of mentions

500
400
300
200
100
0

2020

2021

2022

2023

2024

2025

Year

Nausea

Fatigue

Vomiting

Constipation

Diarrhoea

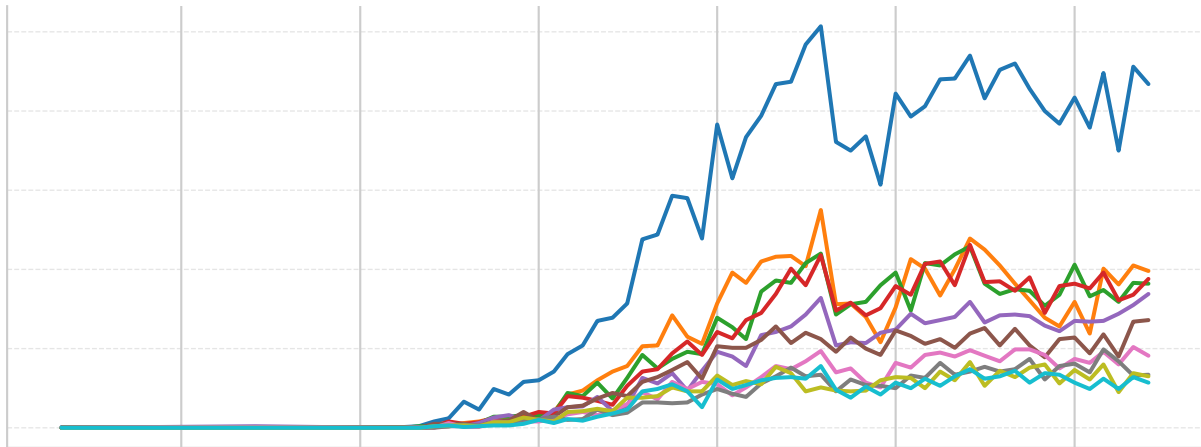
Decreased appetite

Abdominal pain

Eructation

Gastrooesophageal reflux disease

Headache



Appendix Table 1. Co-Occurring PT Frequencies Among 29,172 Reddit Users Taking GLP-1 RAs

For brevity, co-occurring PTs appearing among less than 0.5% of users are not displayed. Users may be counted in multiple PT pairs if they reported more than two symptoms.

PTs	# Users Self Reporting	% of Users with Both PTs
Nausea, Vomiting	2917	10.0
Fatigue, Nausea	2091	7.2
Constipation, Nausea	2039	7.0
Diarrhea, Nausea	1742	6.0
Nausea, Decreased appetite	1592	5.5
Diarrhea, Vomiting	1274	4.4
Abdominal pain, Nausea	1165	4.0
Gastroesophageal reflux disease, Nausea	975	3.3
Constipation, Diarrhea	972	3.3
Eructation, Nausea	968	3.3
Headache, Nausea	952	3.3
Constipation, Fatigue	893	3.1
Abdominal pain, Diarrhea	882	3.0
Constipation, Vomiting	799	2.7
Fatigue, Decreased appetite	771	2.6
Fatigue, Vomiting	769	2.6
Abdominal pain, Vomiting	750	2.6
Diarrhea, Eructation	738	2.5
Abdominal distension, Nausea	702	2.4
Dizziness, Nausea	685	2.4
Vomiting, Decreased appetite	668	2.3
Eructation, Vomiting	652	2.2
Constipation, Decreased appetite	631	2.2
Diarrhea, Fatigue	626	2.2
Abdominal pain, Constipation	606	2.1
Fatigue, Headache	541	1.9
Constipation, Gastroesophageal reflux disease	505	1.7
Diarrhea, Decreased appetite	480	1.7
Constipation, Eructation	465	1.6
Abdominal distension, Constipation	459	1.6
Flatulence, Nausea	459	1.6
Gastroesophageal reflux disease, Vomiting	436	1.5
Abdominal pain, Eructation	431	1.5
Abdominal pain, Fatigue	426	1.5
Fatigue, Gastroesophageal reflux disease	411	1.4
Diarrhea, Flatulence	406	1.4
Dizziness, Fatigue	406	1.4
Eructation, Flatulence	401	1.4

Anxiety, Nausea	384	1.3
Asthenia, Nausea	383	1.3
Abdominal discomfort, Nausea	382	1.3
Headache, Vomiting	382	1.3
Abdominal distension, Diarrhea	376	1.3
Gastrointestinal disorder, Nausea	371	1.3
Constipation, Headache	364	1.3
Dyspepsia, Nausea	356	1.2
Eructation, Fatigue	349	1.2
Diarrhea, Gastroesophageal reflux disease	348	1.2
Abdominal distension, Abdominal pain	336	1.2
Abdominal distension, Eructation	333	1.1
Abdominal pain, Decreased appetite	333	1.1
Insomnia, Nausea	332	1.1
Nausea, Early satiety	331	1.1
Constipation, Flatulence	321	1.1
Abdominal distension, Vomiting	320	1.1
Asthenia, Fatigue	319	1.1
Abdominal distension, Flatulence	317	1.1
Gastroesophageal reflux disease, Decreased appetite	307	1.1
Abdominal distension, Fatigue	302	1.0
Eructation, Decreased appetite	291	1.0
Depression, Fatigue	290	1.0
Eructation, Gastroesophageal reflux disease	283	1.0
Malaise, Nausea	279	1.0
Abdominal pain, Flatulence	276	1.0
Headache, Decreased appetite	275	0.9
Fatigue, Insomnia	267	0.9
Dizziness, Vomiting	265	0.9
Abdominal pain, Gastroesophageal reflux disease	259	0.9
Diarrhea, Headache	259	0.9
Constipation, Dizziness	253	0.9
Anxiety, Fatigue	252	0.9
Flatulence, Vomiting	249	0.9
Alopecia, Nausea	247	0.9
Depression, Nausea	247	0.9
Dehydration, Nausea	239	0.8
Abdominal distension, Decreased appetite	234	0.8
Myalgia, Nausea	228	0.8
Early satiety, Decreased appetite	228	0.8
Hypoglycaemia, Nausea	227	0.8
Nausea, Retching	225	0.8
Dizziness, Decreased appetite	224	0.8

Anxiety, Depression	219	0.8
Nausea, Food aversion	214	0.7
Fatigue, Somnolence	213	0.7
Dyspepsia, Gastrooesophageal reflux disease	211	0.7
Fatigue, Gastrointestinal disorder	210	0.7
Dizziness, Headache	209	0.7
Abdominal discomfort, Diarrhea	206	0.7
Fatigue, Myalgia	204	0.7
Abdominal discomfort, Vomiting	204	0.7
Constipation, Dyspepsia	198	0.7
Hyperhidrosis, Nausea	197	0.7
Abdominal discomfort, Constipation	196	0.7
Fatigue, Flatulence	195	0.7
Diarrhea, Dizziness	193	0.7
Nausea, Somnolence	193	0.7
Abdominal pain, Headache	193	0.7
Diarrhea, Gastrointestinal disorder	193	0.7
Abdominal distension, Gastrooesophageal reflux disease	192	0.7
Asthenia, Vomiting	190	0.7
Dehydration, Vomiting	190	0.7
Abdominal discomfort, Fatigue	190	0.7
Increased appetite, Nausea	189	0.7
Constipation, Gastrointestinal disorder	188	0.6
Dyspepsia, Vomiting	185	0.6
Gastrooesophageal reflux disease, Headache	185	0.6
Nausea, Pain	184	0.6
Alopecia, Fatigue	180	0.6
Dysgeusia, Nausea	180	0.6
Anxiety, Constipation	179	0.6
Fatigue, Malaise	178	0.6
Anxiety, Panic attack	175	0.6
Asthenia, Decreased appetite	171	0.6
Abdominal discomfort, Decreased appetite	171	0.6
Insomnia, Decreased appetite	170	0.6
Gastrointestinal disorder, Vomiting	169	0.6
Nausea, Tachycardia	164	0.6
Constipation, Insomnia	164	0.6
Alopecia, Constipation	163	0.6
Retching, Vomiting	162	0.6
Anxiety, Vomiting	159	0.6
Gastrointestinal disorder, Decreased appetite	158	0.5
Flatulence, Gastrooesophageal reflux disease	154	0.5

Abdominal discomfort, Abdominal pain	153	0.5
Anxiety, Decreased appetite	152	0.5
Flatulence, Decreased appetite	151	0.5
Diarrhea, Dyspepsia	151	0.5
Chills, Nausea	151	0.5
Malaise, Vomiting	150	0.5
Eructation, Headache	150	0.5
Fatigue, Hypoglycaemia	150	0.5
Migraine, Nausea	148	0.5
Food aversion, Decreased appetite	147	0.5
Nausea, Syncope	146	0.5

Appendix Table 2. Frequency of MedDRA Preferred Terms (PTs) among 25,062 Reddit Exclusively Disclosing One Formulation

For brevity, PTs occurring among less than 0.5% of users are not displayed

MedDRA Preferred Term	Semaglutide (n=17,937)		Tirzepatide (n=7,125)	
	n	%	n	%
Nausea	7061	39.37	2039	28.62
Vomiting	3236	18.04	789	11.07
Fatigue	2885	16.08	1046	14.68
Constipation	2670	14.89	918	12.88
Diarrhoea	2229	12.43	888	12.46
Decreased appetite	1999	11.14	847	11.89
Abdominal pain	1524	8.5	571	8.01
Eructation	1188	6.62	502	7.05
Gastrooesophageal reflux disease	1169	6.52	354	4.97
Headache	1078	6.01	409	5.74
Abdominal distension	934	5.21	320	4.49
Dizziness	834	4.65	365	5.12
Anxiety	664	3.7	313	4.39
Flatulence	627	3.5	252	3.54
Abdominal discomfort	548	3.06	194	2.72
Depression	498	2.78	137	1.92
Alopecia	467	2.6	260	3.65
Gastrointestinal disorder	455	2.54	186	2.61
Early satiety	429	2.39	133	1.87
Asthenia	429	2.39	182	2.55
Insomnia	426	2.37	269	3.78
Dyspepsia	381	2.12	152	2.13
Pain	344	1.92	156	2.19
Hypoglycaemia	319	1.78	154	2.16
Malaise	297	1.66	145	2.04
Somnolence	279	1.56	110	1.54
Increased appetite	271	1.51	105	1.47
Tachycardia	264	1.47	139	1.95
Dysgeusia	258	1.44	78	1.09
Dehydration	247	1.38	127	1.78
Retching	232	1.29	79	1.11
Hyperhidrosis	219	1.22	83	1.16
Myalgia	201	1.12	200	2.81
Lethargy	196	1.09	52	0.73
Weight decreased	192	1.07	84	1.18
Weight increased	191	1.06	74	1.04
Tremor	190	1.06	91	1.28
Food aversion	188	1.05	99	1.39
Syncope	188	1.05	95	1.33

Back pain	180	1	96	1.35
Migraine	178	0.99	66	0.93
Arthralgia	166	0.93	89	1.25
Dry mouth	165	0.92	95	1.33
Pruritus	161	0.9	108	1.52
Paraesthesia	160	0.89	98	1.38
Cognitive disorder	160	0.89	58	0.81
Intermenstrual bleeding	156	0.87	76	1.07
Heavy menstrual bleeding	145	0.81	71	1
Irritability	145	0.81	55	0.77
Panic attack	144	0.8	82	1.15
Chills	140	0.78	124	1.74
Injection site pain	135	0.75	81	1.14
Burning sensation	131	0.73	34	0.48
Visual impairment	127	0.71	46	0.65
Acne	123	0.69	36	0.51
Menstruation irregular	114	0.64	61	0.86
Hunger	113	0.63	48	0.67
Dyspnoea	111	0.62	35	0.49
Rash	110	0.61	46	0.65
Taste disorder	106	0.59	41	0.58
Palpitations	104	0.58	57	0.8
Menstrual disorder	101	0.56	40	0.56
Muscle atrophy	100	0.56	50	0.7
Anhedonia	99	0.55	39	0.55
Breath odour	98	0.55	44	0.62
Gastrointestinal motility disorder	96	0.54	36	0.51
Dysmenorrhoea	93	0.52	51	0.72
Dysphagia	93	0.52	31	0.44
Mood swings	92	0.51	29	0.41
Libido decreased	91	0.51	42	0.59
Dermatitis allergic	90	0.5	43	0.6
Contusion	82	0.46	44	0.62
Depressed mood	80	0.45	37	0.52
Impaired gastric emptying	77	0.43	43	0.6
Pyrexia	78	0.43	39	0.55
Muscle spasms	75	0.42	53	0.74
Hot flush	76	0.42	38	0.53
Feeling cold	71	0.4	74	1.04
Hypotension	72	0.4	62	0.87
Sleep disorder	71	0.4	46	0.65
Urticaria	68	0.38	46	0.65
Musculoskeletal pain	57	0.32	41	0.58
Pain in extremity	47	0.26	41	0.58
Neck pain	46	0.26	40	0.56

Injection site reaction	34	0.19	62	0.87
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Appendix Table 3. Frequency of MedDRA Preferred Terms (PTs) among 29,172 Reddit Users Taking GLP-1 RAs Ordered by Frequency

For brevity, PTs occurring among less than 0.5% of users are not displayed

MedDRA Preferred Term	# Users Self Reporting	% of Users with Self-Disclosed PT
Nausea	10764	36.9
Fatigue	4870	16.7
Vomiting	4749	16.3
Constipation	4463	15.3
Diarrhea	3686	12.6
Decreased appetite	3371	11.6
Abdominal pain	2493	8.5
Eructation	2010	6.9
Gastrooesophageal reflux disease	1877	6.4
Headache	1784	6.1
Abdominal distension	1488	5.1
Dizziness	1445	5
Anxiety	1230	4.2
Flatulence	1055	3.6
Alopecia	914	3.1
Insomnia	901	3.1
Abdominal discomfort	900	3.1
Gastrointestinal disorder	883	3
Depression	827	2.8
Asthenia	738	2.5
Early satiety	666	2.3
Dyspepsia	662	2.3
Pain	619	2.1
Hypoglycaemia	600	2.1
Malaise	589	2
Increased appetite	549	1.9
Myalgia	540	1.9
Tachycardia	512	1.8
Somnolence	492	1.7
Dehydration	473	1.6
Weight increased	415	1.4
Dysgeusia	398	1.4
Food aversion	365	1.3
Hyperhidrosis	362	1.2
Retching	350	1.2

Syncope	346	1.2
Arthralgia	336	1.2
Pruritus	335	1.1
Tremor	333	1.1
Chills	332	1.1
Back pain	325	1.1
Dry mouth	321	1.1
Lethargy	311	1.1
Paraesthesia	301	1
Migraine	294	1
Injection site pain	283	1
Panic attack	280	1
Cognitive disorder	267	0.9
Intermenstrual bleeding	266	0.9
Heavy menstrual bleeding	261	0.9
Irritability	257	0.9
Hunger	220	0.8
Visual impairment	218	0.7
Menstruation irregular	211	0.7
Palpitations	205	0.7
Muscle atrophy	194	0.7
Rash	193	0.7
Anhedonia	191	0.7
Burning sensation	190	0.7
Breath odour	187	0.6
Feeling cold	183	0.6
Acne	183	0.6
Menstrual disorder	178	0.6
Taste disorder	177	0.6
Food craving	171	0.6
Dyspnoea	169	0.6
Dysmenorrhoea	168	0.6
Gastrointestinal motility disorder	167	0.6
Impaired gastric emptying	167	0.6
Hypotension	166	0.6
Dermatitis allergic	166	0.6
Libido decreased	161	0.6
Mood swings	159	0.5
Muscle spasms	158	0.5

Contusion	157	0.5
Illness	156	0.5
Hot flush	151	0.5
Urticaria	149	0.5
Apathy	149	0.5
Sleep disorder	148	0.5
Thirst	147	0.5
Injection site reaction	144	0.5
Depressed mood	142	0.5
Dysphagia	142	0.5
Pyrexia	139	0.5
Musculoskeletal pain	134	0.5

Appendix Table 4. Frequency of MedDRA High Level Terms (HLTs) among 29,172 Reddit Users Taking GLP-1 RAs

Only the primary HLT for each Preferred Term (PT) is included, and HLTs reported by fewer than 0.5% of users are excluded.

MedDRA High Level Term	# Users Self Reporting	% of Users with Self-Disclosed HLT
Nausea and vomiting symptoms	12690	43.5
Gastrointestinal atonic and hypomotility disorders NEC	5946	20.4
Asthenic conditions	5677	19.5
Appetite disorders	3979	13.6
Diarrhea (excl infective)	3687	12.6
Gastrointestinal and abdominal pains (excl oral and throat)	2601	8.9
Dyspeptic signs and symptoms	2551	8.7
Flatulence, bloating and distension	2227	7.6
Headaches NEC	1789	6.1
Feelings and sensations NEC	1689	5.8
Neurological signs and symptoms NEC	1508	5.2
Gastrointestinal signs and symptoms NEC	1310	4.5
Anxiety symptoms	1288	4.4
Disturbances in consciousness NEC	1109	3.8
Alopecias	918	3.1
Disturbances in initiating and maintaining sleep	908	3.1
Gastrointestinal disorders NEC	891	3.1
Pain and discomfort NEC	856	2.9
Depressive disorders	827	2.8
Sensory abnormalities NEC	753	2.6
Physical examination procedures and organ system status	721	2.5
Menstruation and uterine bleeding NEC	710	2.4
Musculoskeletal and connective tissue pain and discomfort	641	2.2
Injection site reactions	619	2.1
Hypoglycaemic conditions NEC	602	2.1
Rate and rhythm disorders NEC	548	1.9
Muscle pains	544	1.9
Paraesthesias and dysaesthesias	495	1.7
Total fluid volume decreased	473	1.6
Apocrine and eccrine gland disorders	420	1.4
Muscle related signs and symptoms NEC	395	1.4
General nutritional disorders NEC	392	1.3
Emotional and mood disturbances NEC	383	1.3
Oral dryness and saliva altered	362	1.2
Dermal and epidermal conditions NEC	350	1.2

General signs and symptoms NEC	347	1.2
Joint related signs and symptoms	341	1.2
Pruritus NEC	335	1.1
Tremor (excl congenital)	333	1.1
Mood alterations with depressive symptoms	323	1.1
Migraine headaches	314	1.1
Mental impairment (excl dementia and memory loss)	288	1
Panic attacks and disorders	282	1
Menstruation with increased bleeding	262	0.9
Upper respiratory tract signs and symptoms	240	0.8
Dermatitis and eczema	235	0.8
Visual impairment and blindness (excl colour blindness)	223	0.8
Vascular hypotensive disorders	208	0.7
Cardiac signs and symptoms NEC	205	0.7
Rashes, eruptions and exanthems NEC	203	0.7
Peripheral vascular disorders NEC	201	0.7
Acnes	195	0.7
Inner ear signs and symptoms	190	0.7
Breathing abnormalities	188	0.6
Gastrointestinal dyskinetic disorders	187	0.6
Skin injuries NEC	173	0.6
Sexual desire disorders	171	0.6
Fluctuating mood symptoms	159	0.5
Mood disorders NEC	157	0.5
Parasomnias	156	0.5
Menstruation with decreased bleeding	155	0.5
Urticarias	150	0.5
Sleep disorders NEC	148	0.5
Bladder and urethral symptoms	142	0.5
Febrile disorders	139	0.5

Appendix Table 5. Frequency of MedDRA High Level General Terms (HLGTs) among 29,172 Reddit Users Taking GLP-1 RAs

Only the primary HLGT for each Preferred Term (PT) is included, and HLGTs reported by fewer than 0.5% of users are excluded.

MedDRA High Level General Term	# Users Self Reporting	% of Users with Self-Disclosed HLGT
Gastrointestinal signs and symptoms	15879	54.4
Gastrointestinal motility and defaecation conditions	8576	29.4
General system disorders NEC	7833	26.9
Appetite and general nutritional disorders	4196	14.4
Neurological disorders NEC	3555	12.2
Headaches	2031	7
Skin appendage conditions	1532	5.3
Anxiety disorders and symptoms	1421	4.9
Sleep disorders and disturbances	1140	3.9
Depressed mood disorders and disturbances	1051	3.6
Epidermal and dermal conditions	1005	3.4
Muscle disorders	992	3.4
Menstrual cycle and uterine bleeding disorders	923	3.2
Gastrointestinal conditions NEC	894	3.1
Physical examination and organ system status topics	721	2.5
Musculoskeletal and connective tissue disorders NEC	696	2.4
Mood disorders and disturbances NEC	684	2.3
Electrolyte and fluid balance conditions	630	2.2
Administration site reactions	619	2.1
Glucose metabolism disorders (incl diabetes mellitus)	614	2.1
Cardiac arrhythmias	562	1.9
Salivary gland conditions	363	1.2
Movement disorders (incl parkinsonism)	356	1.2
Joint disorders	354	1.2
Mental impairment disorders	329	1.1
Respiratory tract signs and symptoms	322	1.1
Urinary tract signs and symptoms	306	1
Respiratory disorders NEC	290	1
Vision disorders	252	0.9
Infections - pathogen unspecified	243	0.8
Decreased and nonspecific blood pressure disorders and shock	213	0.7
Vascular disorders NEC	213	0.7
Cardiac disorders, signs and symptoms	207	0.7

NEC		
Injuries NEC	195	0.7
Inner ear and VIIIth cranial nerve disorders	191	0.7
Sexual dysfunctions, disturbances and gender identity disorders	188	0.6
Body temperature conditions	182	0.6
Viral infectious disorders	159	0.5
Angioedema and urticaria	151	0.5

SUPPLEMENTARY METHODS

Subreddits and Keywords used

Subreddits: 'r/mounjaro', 'r/ozempic', 'r/fasting', 'r/intermittentfasting', 'r/keto', 'r/loseit', 'r/Semaglutide', 'r/SuperMorbidlyObese', 'r/PlusSize'

GLP-1 RA drug keywords: "ozempic", "ozempick", "ozenpic", "ozeampic", "ozzempic", "ozempik", "osempic", "ozepmic", "ozeempic", "ozempec", "mounjaro", "mounjoro", "mounjro", "mounjrao", "mounjarro", "monjaro", "mounjard", "moonjaro", "mounjaroo", "munjaro", "semaglutide", "semaglutid", "semaglutite", "semaglutide", "semagltide", "semaglutied", "samaglutide", "semaglutde", "semagultide", "semglutide", "semaglitide", "tirzepatide", "tirzeptide", "tirzepetide", "tirzapatide", "tirzepatid", "trizepatide", "tirzapatiede", "tirzeptid", "tirzepatiede", "tirzepatyde", "wegovy", "wegovi", "weegovy", "wegowvy", "wgovy", "wigovy", "weggovy", "weegovi", "rybelsus", "rybelsos", "ryblesus", "rybeslus", "rybelsu", "ribelsus", "rybelssus", "rybelsis", "ryebلس", "ribelsis", "zepbound", "zepbond", "zepboud", "zepboun", "zebpound", "zepound", "zeppound", "zebound", "zepboudn", "zepboind"

GPT-4o-mini prompt used to determine GLP-1 RA drug usage

System Prompt:

You are an expert researcher looking around reddit for posts/comments

describing self-disclosures of weight loss treatments experienced by the author.

Problem Setting

> You are interested in self-reported effects of a treatment on a user who

took the treatment themselves. You want to be able to answer the following question from the text of the post or comment:

1. Is the user currently take a weight loss medication? Planning to take the medication is not the same as having taken it.

- yes

- no

2. If they are not currently taking it, is the user planning to take a weight loss medication in the future?

- yes

- no

3. If they are not currently taking it, has the user taken a weight loss medication in the past?

- yes

- no

If yes to any of the above:

4. Which medication is being taken (if disclosed)

5. Dose of medication (if disclosed)

6. What is their age (if disclosed)

7. What is their gender (if disclosed)

8. When they started taking the medication. Can be the exact date, number of months, number of weeks, etc. (can be a date in the past if currently taking, date in the past if previously took, or null if not disclosed)

9. When they stopped taking the medication (if previously took the medication)

10. Starting Weight (if disclosed)

11. Current Weight (if disclosed)

12. Weight lost (if disclosed)

13. Other pre-existing medical conditions (if disclosed)

14. Side effects of weight loss medication (if disclosed)

15. Location, e.g. state, city, country (if disclosed)

16. Mental health symptoms before taking the medication (if disclosed)

17. Mental health symptoms while taking the medication (if disclosed)

18. Mental health symptoms after taking the medication (if disclosed)

respond in json format

{

```
"currently_taking_weight_loss_medication",  
  
"planning_to_take_in_future",  
  
"took_in_past",  
  
"medication",  
  
"dose",  
  
"age",  
  
"gender",  
  
"start_date",  
  
"stop_date",  
  
"start_weight",  
  
"current_weight",  
  
"weight_lost",  
  
"pre_existing_conditions",  
  
"side_effects",  
  
"location",  
  
"mh_symptoms_before",  
  
"mh_symptoms_during",  
  
"mh_symptoms_after"  
}
```

Examples

posted in r/Mounjaro

posted on 2023-01-25

So I have been having a hard time finding 5mg in stock until this week, being delivered tomorrow, the last time I took my 5mg shot was on 1/12/23. Do you guys think if I take this shot tomorrow after skipping a week I will be sick?

```
{  
  "currently_taking_weight_loss_medication": "yes",  
  "planning_to_take_in_future": null,  
  "took_in_past": null,  
  "medication": "Mounjaro",  
  "dose": "5mg",  
  "age": null,  
  "gender": null,  
  "start_date": "before 2023-01-12",  
  "start_weight": null,  
  "current_weight": null,  
  "weight_lost": null,  
  "pre_existing_conditions": null,  
  "side_effects": null,  
  "location": null,  
  "mh_symptoms_before": null,  
  "mh_symptoms_during": null,  
  "mh_symptoms_after": null  
}
```

posted in r/Mounjaro

posted on 2023-01-08

I am on Mounjaro 10 mg and have another box of it in my fridge I was able to get with a successful coupon pick up. My insurance now covers Wegovy and I was able to get it. My doctor is having me start at 1.7 mg to stabilize and then go to 2.4mg. Did anyone have any success with weight loss on Wegovy

compared to Mounjaro? I know Mounjaro is the best to get and I would like to use my coupon till I can't anymore but I can't afford the out of pocket cost. SW 231 CW 199 GW 175.

```
{  
  "currently_taking_weight_loss_medication": "yes",  
  "planning_to_take_in_future": null,  
  "took_in_past": null,  
  "medication": "Mounjaro, Wegovy",  
  "dose": "10mg, 1.7mg (starting dose), 2.4mg (future dose)",  
  "age": null,  
  "gender": null,  
  "start_date": null,  
  "stop_date": null,  
  "start_weight": "231",  
  "current_weight": "199",  
  "weight_lost": "32",  
  "pre_existing_conditions": null,  
  "side_effects": null,  
  "location": null,  
  "mh_symptoms_before": null,  
  "mh_symptoms_during": null,  
  "mh_symptoms_after": null  
}
```

User Prompt:

posted in r/{subreddit}\nposted on {date}\n{post or comment text}

Temperature: 0.0

GPT-4.1-mini prompt used to extract side effects

System Prompt:

Extract all self-disclosed side effects from a given social media comment about a GLP-1 drug and match them with the appropriate medDRA concept from the provided knowledge base. Only match side effects that the author of the post is experiencing themselves.

Use the attached medDRA knowledge base to identify and match the appropriate concepts.

Steps

1. ****Identify Side Effects****: Review the comment to extract any stated side effects the individual may have disclosed.
2. ****Match with medDRA Concepts****: Cross-reference the identified side effects with the attached medDRA file to find the matching concept for each extracted side effect.
3. ****Record the Mappings****: List all extracted side effects along with their corresponding medDRA concepts.

Output Format

The output should be in a JSON format containing an array of objects. Each object should include:

- `"side_effect"`: The exact side effect mentioned in the comment.
- `"medDRA_concept"`: The matched medDRA concept from the knowledge base.

Example:

```
```json
```

```
{"side_effect_list":[{"side_effect":"nausea","medDRA_concept":"Gastrointestinal
nausea"}, {"side_effect":"headache","medDRA_concept":"Headache"}]}
```

...

## # Examples

### \*\*Example 1\*\*

- **Input**: "I've been on this GLP-1 drug and I often experience nausea and sometimes headaches."

- **Output**:

```json

```
{"side_effect_list":[{"side_effect":"nausea","medDRA_concept":"Gastrointestinal  
nausea"}, {"side_effect":"headache","medDRA_concept":"Headache"}]}
```

...

Example 2

- **Input**: "Since starting the drug, I can't escape this constant dizziness and fatigue."

- **Output**:

```json

```
{"side_effect_list":[{"side_effect":"dizziness","medDRA_concept":"Dizziness"}, {"side_effect":"fatigue","medDRA_co
ncept":"Fatigue"}]}
```

...

## # Notes

- Return EXACTLY ONE JSON object and nothing else—no explanations, no search log, no second copy.

- Ensure that all side effects mentioned in the comment are examined for potential matches.

- Use the medDRA file provided to ensure complete and accurate concept matching.



- If a side effect does not have an exact match, choose the closest available concept.
- For example "puking" is not a medDRA concept but matches with Vomiting
- If the comment mentions a side effect that someone else experienced, don't return it. Only return side effects that the author themselves had
- If no side effects are present, return {"side\_effect\_list":[]}
- Only return side effects from taking GLP-1 drugs, not side effects from other drugs

Tool Choice: file\_search (vector store of all medDRA preferred terms)

Temperature: 0.0